For ip addresses use the following ranges

+**Vlan 40:** 40.40.40.0/29

+**vlan 70:** 70.70.70.0/29

+**Vlan 80:** 80.80.80.0/29

+**Vlan 90:** 90.90.90.0/29

+Vlan 65: 65.65.65.0/29

+**Vlan 50:** 50.50.50.0/29

+configure hostname on all devices as showing in the graph

##########

R1 and R2:

##########

+R1 must be HSRP active while R2 must be HSRP standby for vlan 70

+R2 must be HSRP active while R1 must be HSRP standby for vlan 40

++ if the active router become down the standby should remain the active even if the previous active returned on

+please configure DHCP server on all needed routers that provide Ip addresses for vlans 70 and 40 "make sure that the pool of ip addresses contains exactly the needed Ip addresses not more than needed"

############

SW1 and SW2:

############

+configure SW1 as VTP server and SW2 as VTP client

+VTP domain name : CCNA

+VTP password : CCNA

+VTP version : 2

+configure portchannel that only allows vlans "1,70,40"

+ make sure that the native vlan is vlan 57 on SW1,SW2

###

R3:

###

Should be configured as NATing router using static NAT; the PC in VLAN90 should use the global ip address **209.1.1.1** to reach the internet

####################

SW3,SW4,SW5,SW6:

####################

+configure port channel between SW3 and SW6

+SW3 must be the root bridge in STP

+if SW3 goes down SW6 must be the second root

+SW3 must be VTP server

+Sw4 and SW6 must be clients

+SW 5 must be transparent

+ PC4 Should be able to reach the network once the interface is up without any delay

###########

R6 and SW7:

###########

+R6 and SW7 must communicate with each other using LOGICAL L3 interface only

+PC8 and PC7 gateways must be configured on the layer 3 switch.
+all routers and Layer 3 switch must run OSPF to achieve connectivity between all networks
+use OSFP areas 1,2, and 3
+configure telnet on Sw7 so it can be accessed only by PC8
+At the end of your configuration do the following tests

+ All PCs must be able to ping R5
+ PC1 must not be able to ping any other PC, it only can reach R5
+ PC7 can’t ping PC8
+ All other devices can ping each other